Assignment 3 - AWS Cloud System Development

Course code and name: COSC2980 - Cloud Computing

Weight: 40% of your final grade

Type: An individual assessment

Feedback mode:

Late work: A university standard late penalty of 10% (= 2 marks) per each working day applies for up to 5 working days that you are late to submit your work for, unless a special consideration has been granted

Learning Objectives Assessed

- CLO 1: Develop and deploy cloud application using popular cloud platforms.
- CLO 2: Design and develop highly scalable cloud-based applications by creating and configuring virtual machines on the cloud and building private cloud.
- CLO 4: Compare, contrast, and evaluate the key trade-offs between multiple approaches to cloud system design, and Identify appropriate design choices when solving real-world cloud computing problems.
- CLO 5: Write comprehensive case studies analysing and contrasting different cloud computing solutions.
- CLO 6: Make recommendations on cloud computing solutions for an enterprise.

Ready for Life and Work

You will learn to architect, automate and manage an entire AWS-based cloud system—integrating compute, storage, networking, IAM and CI/CD pipelines with Infrastructure as Code. These are exactly the capabilities demanded of Cloud Solutions Architects and DevOps Engineers who design secure, scalable platforms for real-world workloads.

Assessment Details

Assignment 03 is worth 40% of your final grade, and is an **individual** assessment. The instructions this assignment can be downloaded from **here** ↓. In addition, some sample solution architecture documents of past offerings can also be downloaded from **here** . These sample documents will help you understand what you will do as part of Assignment 03 and what a successful response to Assignment 03 would look like. Note that the assignment specifications that these reports were originally developed against might have changed, and these are provided as a **reference only**.

[Note:] The evaluation of Assignment 03 will be done entirely demo-based during **Week 10, 11 and 12**.

Support Resources

This assessment requires that you meet RMIT's expectations for academic integrity. More information and advice on how to avoid plagiarism are available in the Getting Started module.

Open **the academic integrity page**.

Additional library and learning resources are available to help with the assessment in this course

Link to **Assignment Support**.

Submission Instruction

Please refer to the detailed guide provided in the Assignment 03 specification that you downloaded earlier.

Submit your solution artifacts into the Canvas anytime before the due deadline (Please refer to *Assignments* -> *Assignment 03* section for the most up-to-date information). Your assessment will be marked during your demo time, and this submission will only be used for keeping records. However, you work will **NOT** be marked until you submit your work to Canvas.

RMIT Electronic Submission of work for assessment

I declare that in submitting all work for this assessment I have read, understood and agree to the content and expectations of the **assessment declaration.**

Rubric

Criteria	Ratings			Pts
Project idea and project formulation with selection of appropriate technologies	**2 pts** **Project idea clearly formulated with selection of appropriate technologies**	**1 pts** **Project idea partially formulated with selection of most technologies being appropriate**	**0 pts** **Project idea not well formulated (with major issues) with selection of most technologies being inappropriate**	2 pts
Skill development in learning new tools and technologies for project completion	**3 pts** **Demonstration of excellent knowledge in learning new tools and technologies for project completion**	**1.5 pts** **Demonstration of good knowledge (with minor issues) in learning new tools and technologies for project completion**	**0 pts** **Demonstration of poor knowledge (with major issues) in learning new tools and technologies for project completion**	3 pts
Appropriate utilization and full implementation of cloud tools/technologies/services in your project	**25 pts** **Appropriate utilization and full implementation of cloud services and/or APIs in your project according to Criteria 1-5.** 32 to >0.0 Pts Appropriate utilization and full implementation of cloud services and/or APIs in your project according to Criteria 1-5.\n 1. You must use the AWS services under the categories of Compute, Containers, Storage, Networking and Content Delivery, Database, and Analytics on the AWS Management Console for your application. **** 2. Each service/API must be 1) fully implemented and automated in your application and 2) assessed as an appropriate selection for your application by the examiner, in order to receive full marks.**** Note: Fully implemented and	**0 pts** **Appropriate AWS services are not utilized in the implementation**		25 pts

Criteria	Ratings				Pts
	automated services/APIs mean the services/APIs should be automatically invoked by your client interface operations/code/other services other than CLI/AWS Console. **** 3. The following fully implemented AWS services are worth 6 marks per type: Elastic Beanstalk, Lambda, API Gateway, ECS, and EMR. **** 4. The other fully implemented AWS services under the categories of Compute, Containers, Storage, Networking and Content Delivery, Database, and Analytics on the AWS Management Console for your application are worth 3 marks per type. Note: The mark calculation for implemented AWS services cannot be iterated, e.g., if you fully implement an Elastic Beanstalk/ECS/EMR/Lambda service that automatically contains an EC2 service and an S3 service, you can only receive 6 marks in total. **** 5. You are encouraged to use third-party APIs external to AWS cloud services. Each type of fully implemented and automated API is worth 2 mark per type. For example, if you fully impellent and automate a Twitter API and a Facebook API, 4 marks will be rewarded. **** Note: In case students choose to integrate third-party APIs, although there is no limit to the number of such APIs they can integrate, only two would be graded.				
Solution Architecture Document - Summary	**0.5 pts** **This segment is well written**	**0.33 pts** **This segment is written**	**0.17 pts** **This segment is**	**0 pts** **This segment is**	0.5 pts

Criteria	Ratings				Pts
	with substantial details and rich visualizations (if required).	with some details and visualizations (if required).	poorly written.	missing	
Solution Architecture Document - Introduction Elaborate i. What are the motivations behind your idea? ii. What it does? - A high level view. iii. Who are the key beneficiaries	**1 pts** **This segment is well written with substantial details and rich visualizations (if required).**	**0.67 pts** **This segment is written with some details and visualizations (if required).**	**0.33 pts** **This segment is poorly written.**	**0 pts** **This segment is missing**	1 pts
Project Report - Related Work Refer some related works similar to your application.	**1 pts** **This segment is well written with substantial details and rich visualizations (if required).**	**0.67 pts** **This segment is written with some details and visualizations (if required).**	**0.33 pts** **This segment is poorly written.**	**0 pts** **This segment is missing**	1 pts
Project Report - System architecture One or more architectural diagram(s) that shows the function and communication between different cloud components used in your project Note: The diagram(s) must be able to clearly show 1) the whole process how each client interface operation invokes the other components of the system, 2) the detailed interactions (e.g. invoking) among all	**5 pts** **This segment is well written with substantial details and rich visualizations (if required).**	**3 pts** **This segment is written with some details and visualizations (if required).**	**1.5 pts** **This segment is poorly written.**	**0 pts** **This segment is missing**	5 pts

Criteria	Ratings				Pts
the components, and 3) the functions of all the components.					
Project Report - System Descriptions explains the purpose of using each of those components	**1 pts** **This segment is well written with substantial details and rich visualizations (if required).**	**0.67 pts** **This segment is written with some details and visualizations (if required).**	**0.33 pts** **This segment is poorly written.**	**0 pts** **This segment is missing**	1 pts
Project Report - Description of dataset/data structure/APIs/sensors	**1 pts** **This segment is well written with substantial details and rich visualizations (if required).**	**0.67 pts** **This segment is written with some details and visualizations (if required).**	**0.33 pts** **This segment is poorly written.**	**0 pts** **This segment is missing**	1 pts
Project Report - References Important references/website links that you use to develop your application.	**0.5 pts** **This segment is well written with substantial details and rich visualizations (if required).**	**0.33 pts** **This segment is written with some details and visualizations (if required).**	**0.17 pts** **This segment is poorly written.**	**0 pts** **This segment is missing**	0.5 pts
					Total Points: 40